Worksheet 4D: Transcendental Functions via Series

What This Worksheet Is About

So far, we've worked with polynomials. But what about functions like e^x, $\sin(x)$, $\cos(x)$, and $\ln(x)$?

Here's the beautiful secret: **These functions can be written as infinite polynomials (series)!**

Once we write them as polynomials, we can differentiate and integrate term by term using the power rule. This lets us *discover* their derivatives rather than just memorizing them.

Part A: The Exponential Function e^x (18 problems)

The Series Definition

The number $e \approx 2.71828...$ is defined as:

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots$$

And the exponential function e^x is:

$$\boxed{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}}$$

Reminder: $n! = n \times (n-1) \times \cdots \times 2 \times 1$, and $0! = 1$ by convention.

Understanding the Series:

1. Calculate the first five terms of the series for $e^1 = e$:

$1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} =$ _____

2. Compare your answer to $e \approx 2.71828$. How close?

3. Write out the first 6 terms of the series for e^x:

$e^x = 1 + x +$ _____ + _____ + _____ + _____ $+ \cdots$

4. Use the first 4 terms to approximate $e^{0.5}$:

$e^{0.5} \approx 1 + 0.5 + \frac{0.25}{2} + \frac{0.125}{6} =$ _____

Actual value: $e^{0.5} \approx 1.6487$. How close?

Differentiating e^x Term by Term

5. Differentiate each term of the series $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$

Term	Derivative
1	
x	
$\frac{x^2}{2!} = \frac{x^2}{2}$	
$\frac{x^3}{3!} = \frac{x^3}{6}$	
$\frac{x^4}{4!} = \frac{x^4}{24}$	
$\frac{x^5}{5!} = \frac{x^5}{120}$	

6. Write out the derivative series you get:

$\frac{d}{dx}e^x = 0 + 1 +$ _____ + _____ + _____ $+\cdots$

7. Compare to the original series. What do you notice?

8. Complete: $\frac{d}{dx}e^x =$ _____

This is remarkable! The exponential function is its own derivative!

Integrating e^x Term by Term

9. Integrate each term of $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$

Term	Integral
1	
x	
$\frac{x^2}{2}$	
$\frac{x^3}{6}$	
$\frac{x^4}{24}$	

10. Write out the integrated series:

$\int e^x\,dx = x + \frac{x^2}{2} +$ _____ + _____ + _____ $+C$

11. Factor out from each term to show this equals $e^x + C$:

Hint: $x = \frac{x^1}{1!}$, $\frac{x^2}{2} = \frac{x^2}{2!}$, etc.

12. Complete: $\int e^x\,dx =$ _____ $+C$

The Derivative and Integral are Inverses

13. Start with $f(x) = e^x$. Differentiate, then integrate:

$f'(x) =$ _____

$\int f'(x)\,dx =$ _____

14. Start with $g(x) = e^x$. Integrate, then differentiate:

$\int g(x)\,dx =$ _____

$\frac{d}{dx}\left[\int g(x)\,dx\right] =$ _____

Chain Rule and Substitution with e^x

15. Using the chain rule: $\frac{d}{dx}e^{2x} = e^{2x} \cdot 2 = 2e^{2x}$

Find: $\frac{d}{dx}e^{3x} =$ _____

16. Find: $\frac{d}{dx}e^{-x} =$ _____

17. Using substitution: $\int e^{2x}\,dx$

Let $u = 2x$, so $du = 2dx$, meaning $dx = \frac{1}{2}du$

$\int e^{2x}\,dx = \int e^u \cdot \frac{1}{2}du = \frac{1}{2}e^u + C =$ _____ $+C$

18. Find: $\int e^{5x}\,dx =$ _____ $+C$

Part B: Sine and Cosine via Series (24 problems)

The Series Definitions

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

Note: x is in **radians**, not degrees!

Understanding the Patterns:

19. The sine series has only _____ powers of x (odd/even).

20. The cosine series has only _____ powers of x (odd/even).

21. Both series alternate signs: $+ - + - \cdots$

22. Use the first 3 terms to approximate $\sin(0.5)$:

$\sin(0.5) \approx 0.5 - \frac{(0.5)^3}{6} + \frac{(0.5)^5}{120} =$ _____

Actual: $\sin(0.5) \approx 0.4794$. How close?

23. Use the first 3 terms to approximate $\cos(0.5)$:

$\cos(0.5) \approx 1 - \frac{(0.5)^2}{2} + \frac{(0.5)^4}{24} =$ _____

Actual: $\cos(0.5) \approx 0.8776$. How close?

Discovering the Derivative of Sine

24. Differentiate each term of $\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \cdots$

Term	Derivative
x	
$-\frac{x^3}{6}$	
$\frac{x^5}{120}$	
$-\frac{x^7}{5040}$	

25. Write the derivative series:

$\frac{d}{dx}\sin(x) = 1 -$ _____ + _____ − _____ $+\cdots$

26. Compare to the cosine series. What is $\frac{d}{dx}\sin(x)$?

$$\boxed{\frac{d}{dx}\sin(x) = \cos(x)}$$

Discovering the Derivative of Cosine

27. Differentiate each term of $\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots$

Term	Derivative
1	
$-\frac{x^2}{2}$	
$\frac{x^4}{24}$	
$-\frac{x^6}{720}$	

28. Write the derivative series:

$\frac{d}{dx}\cos(x) = 0 - x +$ _____ − _____ $+\cdots$

29. Factor out -1 and compare to the sine series:

$\frac{d}{dx}\cos(x) = -\left(x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots\right) =$ _____

$$\boxed{\frac{d}{dx}\cos(x) = -\sin(x)}$$

The Beautiful Cycle

30. Fill in the differentiation cycle:

$\sin(x) \xrightarrow{\frac{d}{dx}}$ _____ $\xrightarrow{\frac{d}{dx}}$ _____ $\xrightarrow{\frac{d}{dx}}$ _____ $\xrightarrow{\frac{d}{dx}}$ _____

31. How many derivatives does it take to get back to $\sin(x)$?

32. What is $\frac{d^4}{dx^4}\sin(x)$? (The fourth derivative)

33. What is $\frac{d^{100}}{dx^{100}}\cos(x)$?

Hint: $100 = 4 \times 25$*, so we go around the cycle 25 times.*

Integrating Sine and Cosine

34. Since $\frac{d}{dx}\sin(x) = \cos(x)$, what is $\int \cos(x)\,dx$?

35. Since $\frac{d}{dx}\cos(x) = -\sin(x)$, what is $\int \sin(x)\,dx$?

Careful with the sign!

36. Verify by differentiating: $\frac{d}{dx}(-\cos(x)) =$ _____

37. Fill in the integration cycle:

$\sin(x) \xrightarrow{\int}$ _____ $\xrightarrow{\int}$ _____ $\xrightarrow{\int}$ _____ $\xrightarrow{\int}$ _____

(Don't forget the $+C$ in practice, but ignore it for the cycle)

Chain Rule and Substitution with Trig

38. Find: $\frac{d}{dx}\sin(2x) =$ _____

39. Find: $\frac{d}{dx}\cos(3x) =$ _____

40. Find: $\frac{d}{dx}\sin(x^2) =$ _____

41. Find: $\int \cos(4x)\,dx =$ _____ $+C$

42. Find: $\int \sin(2x)\,dx =$ _____ $+C$

Part C: The Natural Logarithm $\ln(x)$ (16 problems)

A Different Approach

Unlike e^x, sin, and cos, the series for $\ln(x)$ doesn't converge for all x. But we can use a related series:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad \text{for } -1 < x \leq 1$$

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43. Use the first 4 terms to approximate $\ln(1.5) = \ln(1 + 0.5)$:

$\ln(1.5) \approx 0.5 - \frac{0.25}{2} + \frac{0.125}{3} - \frac{0.0625}{4} =$ _____

Actual: $\ln(1.5) \approx 0.4055$. How close?

44. Use the first 4 terms to approximate $\ln(2) = \ln(1 + 1)$:

$\ln(2) \approx 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} =$ _____

Actual: $\ln(2) \approx 0.6931$. (The series converges slowly here!)

The Derivative of $\ln(x)$

45. Differentiate each term of $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$

Term	Derivative
x	
$-\frac{x^2}{2}$	
$\frac{x^3}{3}$	
$-\frac{x^4}{4}$	

46. Write the derivative series:

$\frac{d}{dx}\ln(1 + x) = 1 - x +$ _____ $-$ _____ $+ \cdots$

47. This is a geometric series! Recall: $1 - x + x^2 - x^3 + \cdots = \frac{1}{1+x}$

So: $\frac{d}{dx}\ln(1 + x) =$ _____

48. Using the chain rule: If $u = 1 + x$, then $\frac{d}{dx}\ln(u) = \frac{1}{u} \cdot \frac{du}{dx} = \frac{1}{1+x} \cdot 1$

For general $\ln(x)$: $\frac{d}{dx}\ln(x) =$ _____

$$\boxed{\frac{d}{dx}\ln(x) = \frac{1}{x}}$$

The Integral of $\frac{1}{x}$

49. We now know: $\int \frac{1}{x}\,dx = \ln|x| + C$

Why the absolute value? Because $\ln(x)$ is only defined for $x > 0$, but $\frac{1}{x}$ exists for $x \neq 0$.

50. Verify: $\frac{d}{dx}\ln|x| = \frac{1}{x}$ for both $x > 0$ and $x < 0$.

Chain Rule and Substitution with ln

51. Find: $\frac{d}{dx}\ln(2x) =$ _____

52. Find: $\frac{d}{dx}\ln(x^2) =$ _____

Two ways: chain rule, or use $\ln(x^2) = 2\ln(x)$ *first*

53. Find: $\frac{d}{dx}\ln(x^2+1) =$ _____

54. Find: $\int \frac{2x}{x^2+1}\,dx =$ _____ $+C$

Hint: Let $u = x^2 + 1$

55. Find: $\int \frac{3x^2}{x^3+5}\,dx =$ _____ $+C$

56. Find: $\int \frac{1}{2x+3}\,dx =$ _____ $+C$

The ln **and** e^x **Connection**

57. Since e^x and $\ln(x)$ are inverse functions:

$\ln(e^x) =$ _____ and $e^{\ln(x)} =$ _____

58. Differentiate $\ln(e^x)$ using the chain rule and verify it equals 1.

Part D: Putting It All Together (18 problems)

Summary Table

Function	Derivative	Integral
e^x	e^x	$e^x + C$
$\sin(x)$	$\cos(x)$	$-\cos(x) + C$
$\cos(x)$	$-\sin(x)$	$\sin(x) + C$
$\ln(x)$	$\frac{1}{x}$	$x\ln(x) - x + C$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\ln\lvert x\rvert + C$

Mixed Practice — Derivatives

59. $\frac{d}{dx}(e^x + \sin(x))$

60. $\frac{d}{dx}(x^2 + \cos(x))$

61. $\frac{d}{dx}(e^x \cdot x)$ — *Product rule!*

62. $\frac{d}{dx}(x \cdot \sin(x))$

63. $\frac{d}{dx}\left(\frac{e^x}{x}\right)$ — *Quotient rule!*

64. $\frac{d}{dx}(e^{\sin(x)})$ — *Chain rule!*

65. $\frac{d}{dx}(\sin(e^x))$

66. $\frac{d}{dx}(\ln(\cos(x)))$

Mixed Practice — Integrals

67. $\int (e^x + x^2)\,dx$

68. $\int (3\cos(x) - 2\sin(x))\,dx$

69. $\int \frac{e^x+1}{e^x}\,dx$ — *Simplify first!*

70. $\int x \cdot e^x\,dx$ — *Integration by parts!*

Let $u = x$, $dv = e^x dx$. Then $du = dx$, $v = e^x$.

$= xe^x - \int e^x\,dx =$ _____

71. $\int x \cdot \cos(x)\,dx$ — *Integration by parts!*

72. $\int e^x \cdot \sin(x)\,dx$ — *This requires parts twice!* (Challenge)

Euler's Formula — The Grand Unification

73. Recall: $e^{ix} = \cos(x) + i\sin(x)$ where $i = \sqrt{-1}$

Using the series for e^x with x replaced by ix:

$e^{ix} = 1 + (ix) + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \cdots$

Calculate the first few powers of i: - $i^1 = i$ - $i^2 = -1$ - $i^3 = -i$ - $i^4 = 1$ - $i^5 =$ _____

74. Separate the real and imaginary parts of e^{ix}:

Real part: $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots =$ _____

Imaginary part: $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots =$ _____

75. This proves: $e^{ix} = \cos(x) + i\sin(x)$

At $x = \pi$: $e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1 + 0 = -1$

So: $e^{i\pi} + 1 = 0$

This is called **Euler's identity** — often called the most beautiful equation in mathematics!

76. Explain in your own words why e, i, π, 1, and 0 appearing in one equation is remarkable.
